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- Write down the gen solution of the ODE $y' = A y$ if A has eigenvalues $\lambda_1 = -1$, $\lambda_2 = -3$, $\lambda_3 = -4$ with matching eigenvectors v_i . Which part of the solution matters most as $t \rightarrow \infty$?

$$y(t) = \sum_{i=1}^3 c_i e^{\lambda_i t} v_i \approx c_1 e^{-t} v_1$$

$\lambda_1 = -1$ slowest decay

- Write down the gen solution of the ODE $y' = A y$ where $A = \begin{pmatrix} -2 & 0 \\ 0 & -3 \end{pmatrix}$ and give the most important part as $t \rightarrow \infty$

$$y(t) = c_1 e^{-2t} v_1 + c_2 e^{-3t} v_2 \approx c_1 e^{-2t} v_1$$

$$\lambda_1 = -2$$

$$\lambda_3 = -3$$

$$v_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

$$v_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

- Write down the gen solution of ODE $y' = A y$ where

$$A = \begin{pmatrix} -1 & 2 & 0 \\ 2 & -4 & 0 \\ 0 & 0 & 0.1 \end{pmatrix} \text{ and give the most important part as } t \rightarrow \infty.$$

$$y(t) = \sum_{i=1}^3 e^{\lambda_i t} v_i \approx c_3 e^{0.1 t} v_3$$

$$\begin{aligned} \lambda_1 &= -5.2 & v_1 &= \begin{bmatrix} 0.5 \\ -0.8 \\ 0 \end{bmatrix} \\ \lambda_2 &= -0.76 & v_2 &= \begin{bmatrix} 0.8 \\ 0.5 \\ 0 \end{bmatrix} \\ \lambda_3 &= 0.1 & v_3 &= \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \end{aligned}$$

$\lambda_3 > 0$
 $e^{\lambda_3 t} \rightarrow \infty$

2. Write down the gen solution of ODE $y' = A y$ where

$$A = \begin{pmatrix} -2 & 2 & 0 \\ 2 & 1 & 0 \\ 0 & 0 & -2 \end{pmatrix} \text{ and give the most important part as } t \rightarrow \infty.$$

$$y(t) = \sum_{i=1}^3 c_i e^{\lambda_i t} v_i \approx c_3 e^{-2 t} v_3$$

$$\begin{aligned} \lambda_1 &= -3 & v_1 &= \begin{bmatrix} -3 \\ -2 \\ 2 \end{bmatrix} \\ \lambda_2 &= -2 & v_2 &= \begin{bmatrix} 0 \\ 2 \\ 1 \end{bmatrix} \\ \lambda_3 &= 2 & v_3 &= \begin{bmatrix} 1 \\ 2 \\ 0 \end{bmatrix} \end{aligned}$$

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- For the ODE $y' = Ay$ where $A = \begin{pmatrix} -3 & -2 \\ -2 & -3 \end{pmatrix}$

In[] := MatrixForm[Eigensystem[$\begin{pmatrix} -3 & -2 \\ -2 & -3 \end{pmatrix}$]]

Out[] := MatrixForm[

$$\begin{pmatrix} -5 & -1 \\ \{1, 1\} & \{-1, 1\} \end{pmatrix}$$

- Write down the gen sol and the important part as $t \rightarrow \infty$.

$$y(t) = c_1 e^{-5t} \begin{bmatrix} 1 \\ 1 \end{bmatrix} + c_2 e^{-t} \begin{bmatrix} -1 \\ 1 \end{bmatrix} \approx c_2 e^{-t} \begin{bmatrix} -1 \\ 1 \end{bmatrix}$$

- Write the gen sol of $y' = Ay$ with $A = \begin{pmatrix} -3 & 2 \\ -2 & -3 \end{pmatrix}$. Describe the solution as $t \rightarrow \infty$.

$$y(t) = c_1 y_1 + c_2 y_2$$

In[] := MatrixForm[Eigensystem[$\begin{pmatrix} -3 & 2 \\ -2 & -3 \end{pmatrix}$]]

Out[] := MatrixForm[

$$\begin{pmatrix} -3+2i & -3-2i \\ \{-i, 1\} & \{i, 1\} \end{pmatrix}$$

$$a+bi = -3+2i$$

$$p+qi = \begin{bmatrix} -i \\ 1 \end{bmatrix}$$

$$a = -3 \quad b = 2$$

$$p = \begin{bmatrix} 0 \\ 1 \end{bmatrix} \quad q = \begin{bmatrix} -1 \\ 0 \end{bmatrix}$$

$$y_1 = e^{at} \begin{pmatrix} \cos(bt)p - \sin(bt)q \\ \sin(bt)p + \cos(bt)q \end{pmatrix}$$

$$y_2 = e^{at} \begin{pmatrix} \sin(bt)p + \cos(bt)q \\ \cos(bt)p - \sin(bt)q \end{pmatrix}$$



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$$a+bi = -2+3i$$

$$a = -2$$

$$b = 3$$

~~$$p+qi = \{1-i, -i, 1\}$$~~

$$p = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}, q = \begin{bmatrix} -1 \\ -1 \\ 0 \end{bmatrix}$$

Write down the gen solution of ODE $y' = Ay$ with A below.

Give the most important part as $t \rightarrow \infty$.

`MatrixForm[Eigensystem[ $\begin{bmatrix} 5 & -10 & -4 \\ 4 & -6 & -1 \\ 2 & -5 & -4 \end{bmatrix}$ ]]]`

`Out[ ]//MatrixForm=`

$$\begin{pmatrix} -2+3i & -2-3i & -1 \\ \{1-i, -i, 1\} & \{1+i, i, 1\} & \{-1, -1, 1\} \end{pmatrix}$$

$$y_1(t) = e^{at} (\cos(bt)p - \sin(bt)q)$$

$$y_2(t) = e^{at} (\cos(bt)q + \sin(bt)p)$$

$$y(t) = c_1 y_1(t) + c_2 y_2(t) + c_3 e^{-t} \begin{bmatrix} -1 \\ -1 \\ 1 \end{bmatrix} \approx c_3 e^{-t} \begin{bmatrix} -1 \\ -1 \\ 1 \end{bmatrix}$$

2. Write down the gen solution of ODE $y' = Ay$ with A below.

Give the most important part as $t \rightarrow \infty$.

$$y(t) = c_1 y_1 + c_2 y_2 + c_3 y_3 \approx c_3 y_3$$

$$y_3 = e^t \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$$

$$a = -1 \quad b = 2$$

$$p = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} \quad q = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}$$

`Eigensystem[ $\begin{bmatrix} -3 & 4 & 0 \\ -4 & 3 & 2 \\ 0 & 2 & -1 \end{bmatrix}$ ] // MatrixForm`

`Out[ ]//MatrixForm=`

$$\begin{pmatrix} -1+2i & -1-2i & 1 \\ \{1+i, i, 1\} & \{1-i, -i, 1\} & \{1, 1, 1\} \end{pmatrix}$$

$$y_1 = e^{at} (\cos(bt)p - \sin(bt)q)$$

$$y_2 = e^{at} (\sin(bt)p + \cos(bt)q)$$

will

$$\begin{pmatrix} 2 & 1 \\ 0 & 2 \end{pmatrix} - 2 \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$$

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$$\begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} \begin{bmatrix} u_1 \\ u_2 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

$$\begin{aligned} 0u_1 + 1u_2 &= 0 \\ 0u_1 + 0u_2 &= 0 \end{aligned}$$

$$\vec{u} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

For the ODE $y' = Ay$ with A below. Write the gen sol and the important part as $t \rightarrow \infty$.

`in[ ] := MatrixForm[Eigensystem[{{2, 1}, {0, 2}}]]`

`Out[ ]: Eigensystem[{{2, 1}, {0, 2}}]`

$$\begin{pmatrix} 2 & 1 \\ 0 & 2 \end{pmatrix}$$

$$y_1 = e^{2t} \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

$$y_2 = te^{2t} \begin{bmatrix} 1 \\ 0 \end{bmatrix} + e^{2t} \vec{u}$$

$$y(t) = c_1 y_1(t) + c_2 y_2(t)$$

$$= c_2 y_2$$

1. For the ODE $y' = Ay$ with A below. Write the gen sol.

$$y(t) = c_1 e^t \begin{bmatrix} 1 \\ 1 \end{bmatrix} + c_2 y_2$$

missing evec

$$y_2 = te^t \begin{bmatrix} 1 \\ 1 \end{bmatrix} + e^t \vec{u}$$

$$[A - (1)I] \vec{u} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

$$\begin{bmatrix} 1 & -1 & 1 & 1 \\ 1 & -1 & 1 & 1 \end{bmatrix} \sim \begin{bmatrix} 1 & -1 & 1 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\Rightarrow \vec{u} = \begin{bmatrix} 1 \\ 0 \end{bmatrix} + u_2 \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

easy solution is $u_2 = 0$

$$\vec{u} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

`in[ ] := MatrixForm[Eigensystem[{{2, -1}, {1, 0}}]]`

`Out[ ]: Eigensystem[{{2, -1}, {1, 0}}]`

$$\begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix}$$

■ For the ODE $y' = Ay$ with A below. Write the gen sol and the important part as $t \rightarrow \infty$.

$$\text{MatrixForm}[\text{Eigensystem}\left[\begin{pmatrix} 0 & -5 & -2 \\ 1 & 0 & 1 \\ -2 & 4 & 0 \end{pmatrix}\right]]$$

$$\left\{ \begin{pmatrix} 2 \\ -1, 0, 1 \end{pmatrix}, \begin{pmatrix} -1 \\ -1, -1, 2 \end{pmatrix}, \begin{pmatrix} -1 \\ 0, 0, 0 \end{pmatrix} \right\}$$

$$y_1 = e^{2t} \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix}$$

$$y_2 = e^{-t} \begin{bmatrix} -1 \\ -1 \\ 2 \end{bmatrix}$$

$$y_3 = t y_2 + e^{-t} \begin{bmatrix} u_1 \\ u_2 \\ u_3 \end{bmatrix}$$

$$y(t) = c_1 y_1 + c_2 y_2 + c_3 y_3$$

$$= c_1 y_1$$

$$\begin{cases} u_1 = -1 \\ u_2 = 0 \\ u_3 = 0 \end{cases}$$

$$\left\{ A = (-1) \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \right\} \begin{bmatrix} u_1 \\ u_2 \\ u_3 \end{bmatrix} = \begin{bmatrix} -1 \\ -1 \\ 2 \end{bmatrix} \quad \begin{bmatrix} 1 & -5 & -2 \\ 1 & 1 & 1 \\ -2 & 4 & 1 \end{bmatrix} \begin{bmatrix} u_1 \\ u_2 \\ u_3 \end{bmatrix} = \begin{bmatrix} -1 \\ -1 \\ 2 \end{bmatrix}$$

$$\sim \begin{bmatrix} 1 & 0 & \frac{1}{2} & -1 \\ 0 & 1 & \frac{1}{2} & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

2. For the ODE $y' = Ay$ with A below. Write the gen sol and the important part as $t \rightarrow \infty$.

$$y(t) = c_1 y_1 + c_2 y_2 + c_3 y_3$$

$$= c_3 y_3$$

$$y_3 = e^{-t} \begin{bmatrix} -2 \\ 1 \\ 0 \end{bmatrix}$$

$$y_1 = e^{-3t} \begin{bmatrix} 3 \\ -2 \\ 1 \end{bmatrix}$$

$$y_2 = t y_1 + e^{-3t} u$$

$$\text{MatrixForm}[\text{Eigensystem}\left[\begin{pmatrix} 2 & 6 & -3 \\ -2 & -5 & 2 \\ -1 & -2 & -4 \end{pmatrix}\right]]$$

$$\left\{ \begin{pmatrix} -3 \\ 3, -2, 1 \end{pmatrix}, \begin{pmatrix} -3 \\ 0, 0, 0 \end{pmatrix}, \begin{pmatrix} -1 \\ -2, 1, 0 \end{pmatrix} \right\}$$

$$y_2 = t e^{-3t} \begin{bmatrix} 3 \\ -2 \\ 1 \end{bmatrix} + e^{-3t} \begin{bmatrix} -1 \\ 2 \\ 0 \end{bmatrix}$$

$$\left[A - (-3)I \quad \begin{bmatrix} 3 \\ -2 \\ 1 \end{bmatrix} \right]$$

$$\left[\begin{array}{ccc|c} 5 & 6 & -3 & 3 \\ -2 & -2 & 2 & -2 \\ -1 & -2 & -1 & 1 \end{array} \right] \sim \left[\begin{array}{ccc|c} 5 & 6 & -3 & 3 \\ 1 & 1 & -1 & 1 \\ 1 & 2 & 1 & -1 \end{array} \right]$$

$$\sim \left[\begin{array}{ccc|c} 5 & 6 & -3 & 3 \\ 1 & 1 & -1 & 1 \\ 1 & 2 & 1 & -1 \end{array} \right]$$

$$\sim \left[\begin{array}{ccc|c} 1 & 1 & -1 & 1 \\ 5 & 6 & -3 & 3 \\ 1 & 2 & 1 & -1 \end{array} \right] \sim \left[\begin{array}{ccc|c} 1 & 1 & -1 & 1 \\ 0 & 1 & 2 & 2 \\ 0 & 1 & 2 & -2 \end{array} \right]$$

$$\sim \left[\begin{array}{ccc|c} 1 & 1 & -1 & 1 \\ 0 & 1 & 2 & 2 \\ 0 & 0 & 0 & 0 \end{array} \right]$$

choose $u_3 = 0$

$$u_2 = 2$$

$$u_1 = 1 - u_2 = -1$$

